Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/16/2008
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	08/01/2006	**End Date:**	11/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Due to NASA cuts and the decision to remove the research from Flight Status, funding for the co-PIs Robert Hyers and Anup Gangopadhyay was cut and they will be removed as PIs.		
COI Name (Institution):	Herlach, Dieter Ph.D. (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation) Gangopadhyay, Anup (Washington University) Hyers, Robert Ph.D. (University of Massachusetts)		
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Task Description:	In 1724, Fahrenheit first observed that liquid water could be cooled by a significant amount below its freezing temperature before crystalline ice forms. In 1952, David Turnbull showed that this ability to supercool was a common property of liquids, occurring even in metallic liquids. His experiments demonstrated that a barrier exists to the initial formation of the crystal from the liquid, a process called nucleation. This process is the starting point of not only crystallization, but of most phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. The attainment of an improved ability to control nucleation is of central importance for the development of new materials for advanced applications. In almost all cases the classical theory of nucleation, first developed in the late 19th century, is used to analyze and predict nucleation results. There are, however, serious problems with aspects of that theory in significant regimes of its application. Further, it does not properly describe nucleation when the concentrations of the initial and final phases are different. The purpose of the flight experiment QUASI was to provide a benchmark test of the classical theory and of an extension to that theory to account for different compositions of the initial and final phases. To do this, measurements of the fundamental thermodynamic and kinetic parameters of the supercooled liquids and solid crystallization products are required at the same time that the nucleation rates are measured. Unavoidable convective stirring and the inability to make all of the physical measurements under terrestrial gravity make the microgravity environment of the International Space Station (ISS) essential for such studies. The planned flight experiments would have used the Materials Science Laboratory – Electromagnetic Levitator (MSL-EML) currently being developed by the European Space Agency (ESA). As a bonus, this facility would also have allowed the growth rates of the complex crystal and ordered noncrystallographic phases that form to be measured, allowing theoretical improvements of growth processes, also essential for improved microstructural control. The results of these studies would have lead to an improved ability to develop new materials and phase microstructure and to better tune them to desired applications. As we have already demonstrated in our ground-based studies leading to the flight experiments, new techniques and technologies would also have emerged that would have impacted other areas of science and technology. In 2003, my research group at Washington University, St. Louis, MO, teamed with Dr. Jan Rogers and her team at NASA Marshall Space Flight Center, Huntsville, AL, and Dr. Doug Robinson at the MUCAT beamline of the Advanced Photon Source developed a new technique (Beamline Electrostatic Levitation, or BESL) to obtain rapid structural information at high temperature from electrostatically-levitated samples of Ti-Zr-Ni. We demonstrated that icosahedral short-range ordering in the supercooled liquid catalyzed the formation of a metastable quasicrystal, a noncrystallographic ordered phase that is characterized by long range icosahedral order. In addition to verifying for the first time a half-century old hypothesis about nucleation from a liquid, those studies also demonstrated that this technique could be used to rapidly determine the phase diagrams of high temperature materials. This would be of central interest to the development of new materials for advanced propulsion for aerodynamic and space applications. For flight studies of nucleation, it is essential to first understand all aspects of the nucleation process that are accessible with ground based experiments. As mentioned, the ground-based ESL experiments demonstrated a new connection between structure and the nucleation barrier. In addition, they suggested the occurrence of a metastable critical point in the supercooled liquid, a manifestation of the developing icosahedral order. Evidence suggests a coupling between this higher order liquid ordering transition and the nucleation barrier for the ordered phase. This is a previously unknown type of nucleation, not describable by the classical theory of nucleation and possibly impacting the planned nucleation studies on the ISS. The task of the continuing ground-based research on this contract is the investigation of the icosahedral ordering and associated thermophysical properties in supercooled liquids, primarily of transition metals and their alloys. Measurements are being made of the structure, specific heat, density, viscosity, surface tension and emissivity in the liquids. These data, when combined with the nucleation data, will lead to a better understanding of the complicated nucleation processes that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
	In 2005 NASA decided to put on hold planned ISS experiments, including our flight experiment, QUASI. We were allowed a phase-out period for this contract, NNM04AA01G. However, recognizing the importance and visibility of our ground-based research, and the potential for a successful ISS experiment should a future opportunity arise, we were issued a new contract, NNX07AK27G to continue our ground-based studies. The analysis of existing thermophysical and structural data for supercooling liquids obtained at the Advanced Photon Source was supported by residual funds in NNM04AA01G. The results from the past year are summarized below. Some of these studies were also relevant to related work under our NSF grants, DMR 0307410 and DMR 0606065, studying the relations between structure and ordering in the liquid and the glass and the influence of this ordering on glass formation, the glass transition and the crystal nucleation barrier. 1. Previously, we demonstrated the development of icosahedral short-range order (ISRO) in equilibrium and supercooled Ti-Zr-Ni liquids. In-situ high-energy x-ray diffraction data were obtained from electrostatically-levitated liquids using the Beamline Electrostatic Levitation (BESL) method that was developed under this contract. Coupled with nucleation data on the same liquid, these studies were the first demonstration of a half-century hypothesis relating the liquid structure to the nucleation barrier. Since the last report, we have extended those studies to show that from a cluster-based analysis of the X-ray structure factors the ISRO in the alloys becomes progressively more dominant, and the coherence length of the order becomes longer, with the addition of Ni, especially near a concentration of 21 at.% Ni. 2. Undercooling experiments on a large number of simple crystalline, polytetrahedral, and icosahedral quasicrystalline phase forming compositions in Ti-Zr-Ni alloys were carried out using electrostatic levitation (ESL) techniques for containerless processing. Consistent with Frank's hypothesis, a direct correlation was found between the reduced undercooling and the icosahedral order in the solid. The reduced undercooling is less for liquids that form the icosahedral quasicrystal (i-phase) than for those that form the hcp C14 Laves polytetrahedral phase. For many compositions near 21 at.% Ni, the primary nucleation of a metastable i-phase instead of a stable C14 Laves phase

Task Progress:	demonstrates that the interfacial free energy between the liquid and the i-phase is smaller than between the liquid and the C14 Laves phase, indicating icosahedral local order in the undercooled liquid. This is in agreement with a classical-nucleation-theory-based estimate of the interfacial free energy and the work of formation of the critical cluster from the undercooling data. 3. Our earlier conclusions about ISRO were based on the assumption of a single dominant cluster type in the liquid and the observed evolution of a high-q shoulder on the second peak in the structure factor, S(q). New diffraction data for Ti37.5Zr37.5Ni21 were obtained under the support of this contract, using more rapid data acquisition methods. These allowed structural studies to be made down to and through recalescence to the icosahedral quasicrystal. The liquid atomic structures obtained from a Reverse Monte Carlo analysis of these data were characterized by their bond angle distributions, Honeycutt and Andersen indices and bond orientational order parameters. The studies showed that while a single cluster was not able to describe the data, the dominant order was icosahedral, supporting our earlier conclusions. 4. Earlier studies by us, assuming a dominant cluster in the liquid, indicated that the ISRO in early transition liquids (Ti and Zr) was distorted, compared to that in late transition metal liquids such as Ni. To check this further, the atomic structures of the liquids were determined from a RMC analysis of the diffraction data. These were analyzed in terms of their bond angle distributions, Honeycutt-Anderson indices and bond orientational order parameters. Contrary to our previous single-cluster studies these results indicate that there is clearly no single cluster type that fully describes these liquids. However, in limited agreement with the previous cluster-based studies of the diffraction data, supercooled liquid Ti develops a significant (though not exclusive) amount of distorted icosahedral short-range order with supercooling, while the short-range order in supercooled liquid Zr and Ni is less distorted. The short-range order in Ni is particularly close to that of an undistorted icosahedron. This difference in the evolution of the local structures of the early and late transition metals likely reflects a competition between close packing and the angular dependence of the atomic potential. 5. The local structure of supercooled liquid Zr67Pd33 was studied using BESL The distinct shoulder on the high-q side of the second peak in the structure factor indicated the development of ISRO with supercooling. The atomic structures obtained by a Reverse Monte Carlo simulation of the experimental data were analyzed in terms of their Honeycutt and Andersen indices and bond orientational order parameters (BOO) to confirm this. These studies indicate that Zr-centered icosahedral clusters exist in the supercooled liquid and that the fraction of the icosahedral clusters increases with supercooling. 6. Liquid silicon is known to be metallic while the diamond cubic solid form is a well-known semiconductor. Theoretical treatments have argued for a liquid/liquid phase transition in the supercooled state at which the local order changes from the metallic A5 type to the semiconductor A4 one. To investigate this prediction, we have constructed the atomic structures of supercooled liquid silicon from BESL X-ray diffraction data obtained from the levitated liquid. A Reverse Monte Carlo method was used to determine the atomic structures as a function of supercooling. Based on a Honeycutt-Anderson and bond orientational analyses of these structures, the supercooled liquid silicon maintains the A5 structure (with some A4), with the coherence length of the A5 order increasing with supercooling. The ordering of the A5 structure is consistent with previous experimental observations, such as a constant coordination number, constant nearest-neighbor distances and the evolution of the first peak in S(q) and of the second peak in g(r) with supercooling. These observations are in opposition to the theoretical predictions of a liquid-liquid structural phase transition toward a diamond cubic structure (characteristic of crystal Si) in supercooled liquid Si, Further, they indicate that baring the intervention of phase transitions such as crystallization, liquids will order towards a preferred structure that is characteristic of the liquid state and the nature of the atomic bonding.

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